

JEE MAIN REVISION MASTER: ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 178 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Aldehydes Ketones And Carboxylic Acids - Core Application	178 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Aldehydes Ketones And Carboxylic Acids - Core Application (Very High Yield — 178 PYQs)

Core Principle: Key problem pattern in Aldehydes Ketones And Carboxylic Acids.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Consider the following reaction. The major product(P)formed is :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Complete the following reaction sequence and give the name of major product 'P'...." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "A molecule (X) with following structure under mild acidic condition is hydrolysed to produce (Y) and (Z). Identify the correct statements ab..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 22nd January Morning Shift (Q#3)

Match the LIST-I with LIST-II List-I Reagents List-II Name of Reaction involving carbonyl compounds A.

$\text{NH}_2\text{-NH}_2$, KOH I. Tollen's Test B. $\text{Ag}(\text{NH}_3)_2\text{OH}$ II. Clemmensen Reduction C. Aq. CuSO_4 , Sodium Potassium tartarate, KOH III. Wolff - Kishner Reduction D. Zn-Hg, HCl IV. Fehling's Test Choose the correct answer from the options given below :

- (A) -IV, B-III, C-II, D-I
- (B) -III, B-I, C-IV, D-II **[Correct]**
- (C) -II, B-I, C-IV, D-III
- (D) -III, B-IV, C-I, D-II

Correct Answer: B

Solution Analysis: A reagent is NH_2NH_2 , KOH (hydrazine in strong base). This is used to reduce aldehydes/ketones to hydrocarbons by Wolff–Kishner reduction . So, A $\rightarrow$ III . B reagent is $\text{Ag}(\text{NH}_3)_2\text{OH}$ (ammoniacal silver nitrate). This gives silver mirror with aldehydes: Tollen's test . So, B $\rightarrow$ I . C reagent is aqueous CuSO_4 + sodium potassium tartrate (Rochelle salt) + KOH ; together they form Fehling's solution , used in Fehling's test for aldehydes (red Cu_2O ppt). So, C $\rightarrow$ IV . D reagent is Zn - Hg, HCl which reduces carbonyl group to $-\text{CH}_2-$ in ac